

Session 7: Metadata-Driven Pipelines from SFTP to Lakehouse

17 August 2026 20:39

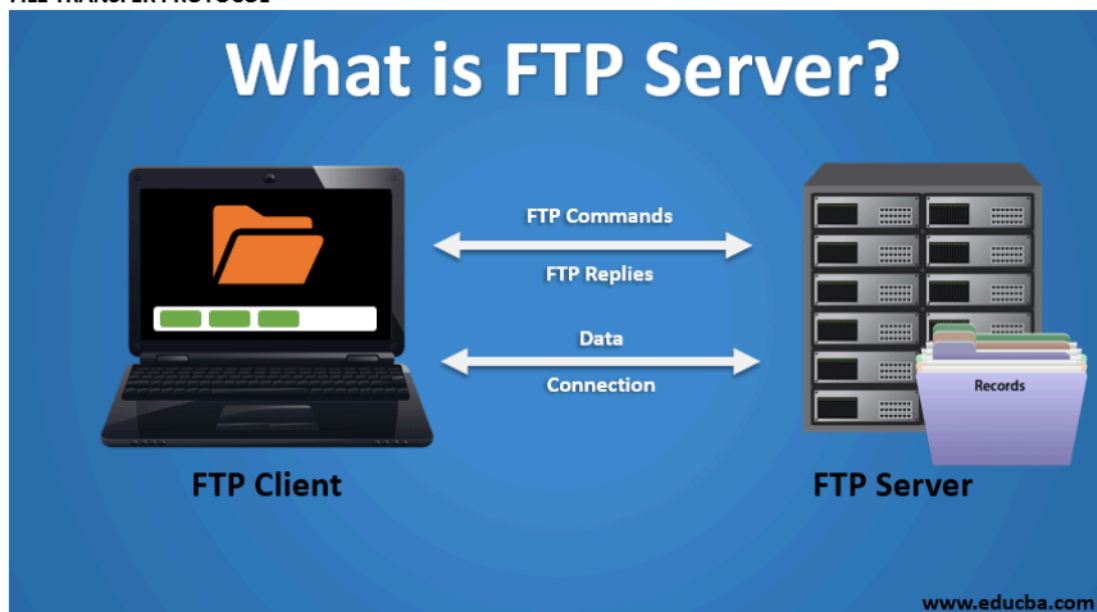
1. FTP vs SFTP

SR.NO	FTP	SFTP
1	File Transfer Protocol	Secure File Transfer Protocol
2	Data is not encrypted	Data is encrypted
3	Less secure	More Secure
4	Uses traditional file transfer	Uses SSH encryption
5	Not recommended for sensitive data	Suitable for secure file transfer

SFTP advantages:

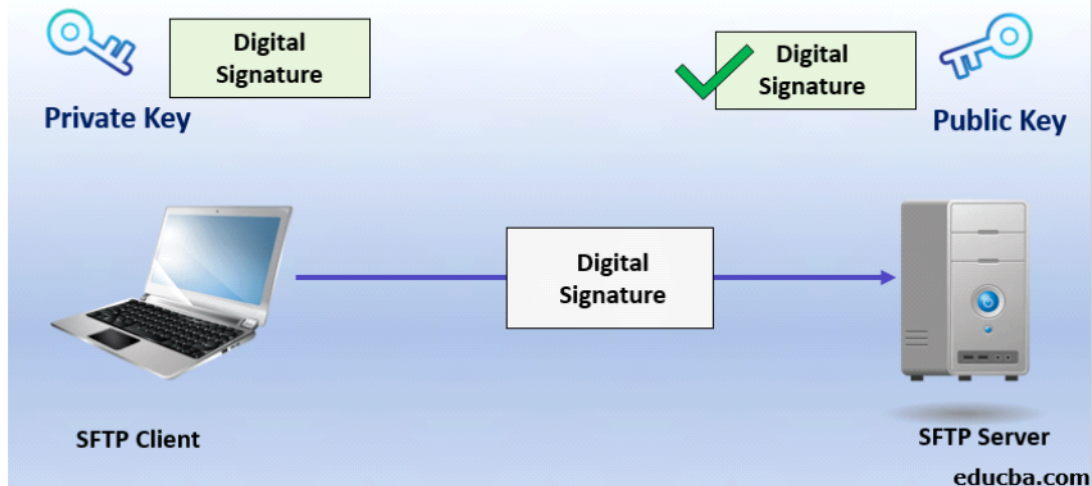
- SSH encryption
- Data encryption
- User and server authentication
- Secure file transfer
- Supports legacy systems
- Commonly uses port 22

FILE TRANSFER PROTOCOL



SECURE FILE TRANSFER PROTOCOL

What is SFTP



2. Why Organizations Use SFTP

- Secure file transfer
 - Simple and familiar
 - Supports legacy systems
 - File-level access control
 - Easy firewall configuration
 - Server-to-server transfers
 - Resumable transfers
- **Use SFTP:** Secure, file-based and legacy integrations.
 - **Use Azure Storage:** Large-scale cloud storage, scalability and integration with Azure services.

3. Business Scenario

Vendors
↓
SFTP Server
↓
Microsoft Fabric Pipeline
↓
Lakehouse Bronze

Typical files:

- Customer
- Sales
- Orders
- Product
- Bank transactions
- Insurance • Payroll

4. Metadata-Driven Pipeline

Hardcoded Approach

- Customer Pipeline
- Employee Pipeline
- Orders Pipeline
- Product Pipeline

Problems:

- Multiple pipelines
- Hardcoded paths
- More maintenance

Metadata-Driven Approach

Metadata Table



One Pipeline



Lookup



ForEach



Copy Activity

Benefits:

- One reusable pipeline
- Configuration driven
- Dynamic paths
- Easy onboarding
- Less maintenance

WHY VENDORS UPLOAD DATA TO SFTP SERVERS

1. ITS EASY TO UPLOAD
2. DUE TO SECURITY PURPOSE
3. SOME OF THE ON-PREMISES SOURCES NOT ABLE TO CONNECT DIRECTLY TO THE CLOUD SO THEY WILL UPLOAD DATA TO SFTP AND FROM SFTP DATA ENGINEERS WILL COLLECT THE DATA

WHEN TO USE SFTP

1. 10 GB DATA WE CAN STORE IN SFTP SERVER

WHEN TO USE ADLS GEN2

1. STORE LARGE DATA - NO LIMIT

THERE ARE TWO WAYS TO CONNECT SFTP

2. FILEZILLA LINK - https://filezilla.en.softonic.com/download#google_vignette

3. WINSCP LINK - <https://winscp.net/eng/download.php>

Password :- HMLkLIIQ28P20g6al17FQgYU8i9nuhp5

Connection string :- adlssftp1808.intellibiuser01@adlssftp1808.blob.core.windows.net

BEFORE @ - USERNAME

AFTER @ - SERVERNAME/ HOST NAME

PORT - 22

Ravindra user

PASSWORD :- FA6eVElObXS4vZigtHvilZYHWKvFR+XI

CONNECTION STRING :- adlssftp1808.ravindra001@adlssftp1808.blob.core.windows.net

```
create table tblconfig(
id int,
src_container varchar(30),
src_folder varchar(30),
src_filename varchar(30),
sink_folder varchar(30),
sink_filename varchar(30)
);
select * from tblconfig;
insert into tblconfig VALUES
(1,'global','customer','cust.csv','cust-out','cust.csv'),
(2,'global','employees','employees.csv','emp-out','emp.csv'),
(3,'global-new','sample','dept.csv','sample-out','dept.csv')
```

```
id
src_container
src_folder
src_filename
sink_folder
sink_filename
```

```
{
  "count": 3,
  "value": [
    {
      "id": 1,
      "src_container": "global",
      "src_folder": "customer",
      "src_filename": "cust.csv",
      "sink_folder": "cust-out",
      "sink_filename": "cust.csv"
    },
    {
      "id": 2,
      "src_container": "global",
      "src_folder": "employees",
      "src_filename": "employees.csv",
      "sink_folder": "emp-out",
      "sink_filename": "emp.csv"
    },
    {
      "id": 3,
      "src_container": "global-new",
      "src_folder": "sample",
      "src_filename": "dept.csv",
      "sink_folder": "sample-out",
      "sink_filename": "dept.csv"
    }
  ]
}
```